

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0986

Rocktenn Coshocton Mill 230 kV
March 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0986. The Interconnection Customer proposes a 224.2 MW wind facility interconnecting at the Rocktenn Coshocton Mill 230 kV substation in the Ohio area. The request was received on May 03, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 286 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,110,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MYRTLE-DATE / WATERFORD POWER LLC 230 KV line	TIDD / MUSKINGUM RIV 500 KV	129.0	637	7.8
AES / KNOX 230 KV	ATASCOCITA / N CAMBRIDGE 230 KV	114.3	468	5.3
FILER 500/345 kV Transformer Bank	LERDO / TIDD 500 KV	106.2	967	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WATERFORD POWER LLC SUBSTATION / MYRTLE-DATE 230 KV	Rebuild line with higher-capacity conductor	\$23,879,000	34% (\$8,123,000)
AES / KNOX 230 KV	Rebuild line with higher-capacity conductor	\$27,021,000	39% (\$10,520,000)
FILER 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,663,000	8% (\$2,467,000)
	Total allocated to project		\$21,110,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.